

Chapter 7: Electronic Spreadsheet

7.1 Introduction to Spreadsheets

Electronic Spreadsheets consist of multiple rows and columns, used to perform simple and complex calculations, present data in charts, sort data, and filter information.

Examples include Microsoft Excel, Apple Numbers, LibreOffice Calc, and OpenOffice Calc.

Interface Components

- **Active Cell:** The currently selected cell, indicated by a thick border.
- **Name Box:** Displays the address of the active cell.
- **Formula Bar:** Displays the contents or the formula of the active cell.
- **Columns & Rows:** Columns are named with letters (A, B, C... AA, AB), and rows are numbered (1, 2, 3...).

7.2 Cell Addresses and Ranges

Cell Address: Denoted first by the column letter and secondly by the row number (e.g., D3).

Range of Cells: A block of adjacent cells. It is represented by the starting cell and ending cell separated by a colon (e.g., B2:B5 or A3:C3).

7.3 Entering Data

The contents of a cell can be a label, a value, or a formula:

- **Labels:** Text or combinations of letters and numbers. By default, they are left-aligned.
- **Values:** Numerical figures. By default, they are right-aligned.
- **Formulae:** Expressions beginning with an equals sign (=) that use values, cell addresses, and operators.

7.4 Mathematical Operators and Precedence

Operator	Meaning
+	Addition
-	Subtraction
*	Multiplication
/	Division
^	Exponentiation (Power)

Order of Evaluation: 1) Brackets (), 2) Exponentiation ^, 3) Multiplication * and Division /, 4) Addition + and Subtraction -.

7.5 Functions

Functions simplify complex calculations over large ranges of data.

- **SUM(range):** Adds the values.
- **AVERAGE(range):** Calculates the average.
- **MAX(range):** Finds the largest value.
- **MIN(range):** Finds the smallest value.
- **COUNT(range):** Counts the number of cells containing numerical values.

7.6 Cell References

- **Relative Reference (e.g., A1):** Row and column change when the formula is copied.
- **Absolute Row Reference (e.g., A\$1):** Row remains unchanged when copied.
- **Absolute Column Reference (e.g., \$A1):** Column remains unchanged when copied.
- **Absolute Reference (e.g., \$A\$1):** Both row and column remain completely unchanged when copied.

7.7 Charts

Charts make data easier to comprehend and analyze.

- **Column Chart & Bar Chart:** Compare classes of data items in a group.
- **Line Chart:** Group comparison over time.
- **Pie Chart:** Compare classes of data items as a percentage of a whole.
- **XY Scatter Chart:** Compare data in pairs.